Shaheed Zulfiqar Ali Bhutto Institute of Science & Technology University

DEPARTMENT OF ROBOTICS & ARTIFICIAL INTELLIGENCE

Total Marks: 7.5

Obtained Marks: _____

Database Management System (LAB)

Project Report

Submitted To: Mr. Annas Khalid Khan

Name	**Registration No.**
Syeda Nalain Zainab Bukhari	24108233
Tehniat Khan	24108235
Aiman Nasir	24108202
Malika Naz	24108212
Safia Alam	24108229

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Table of Contents

1. **Book Library Management System**
   1.1 Introduction
   1.2 Project Background
   1.3 Project Objectives
   1.4 Expected Benefit
2. **Entity Relationship Diagram**
   2.1 Entities
   2.2 Attributes
   2.2.1 Member
   2.2.2 Author
   2.2.3 Publisher
   2.2.4 Book
   2.2.5 Book_Author
   2.2.6 Loan
   2.2.7 Fines
3. **Relationship**
   3.1 One-to-Many Relationships
   3.2 Many-to-Many Relationships
4. **Entity Relationship Diagram ERD (DRAW.IO)**
5. **Member**
6. **Running the Application**
7. **Project Report**
   7.1 Files Imported in Project
   7.2 Tables Created in MySQL
   7.3 Software Tools Used
   7.3.1 Front End
   7.3.2 Back End
8. **Common Issues & Solutions**
9. **Additional Features You Can Add**
10. **Conclusion**
11.

1.Book Library Management System

1.1 INTRODUCTION:

The book library manager is a comprehensive database system design to modernize and automate the core operation of a library.In today's digital age, manual library management systems are prone to errors, inefficient in tracking resources, and inadequate for providing meaningful insights into library operations. This database system aims to replace paper-based records and disjointed spreadsheets with a centralized, reliable, and scalable solution that can handle the complex relationships between books, authors, members, and borrowing activities.It is designed to accommodate the needs of small to medium-sized libraries while being scalable enough for larger institutions.

1.2 Project Background

Libraries today face significant challenges in managing their operations efficiently.Many still rely on manual systems using paper records, index cards, or basic spreadsheets. These methods are time-consuming, prone to errors, and cannot handle the complex relationships between books, authors, members, and borrowing activities effectively.

1.3 Project Objectives

· Create a centralized digital database to replace paper-based systems

. Automate book tracking and member management

· Provide real-time information on book availability

· Streamline the borrowing and return processes

· Generate accurate reports for better decision-making

· Reduce administrative workload through automation

· Improve service quality for library members

1.4 Expected Benefit:

· Faster and more accurate book searches

· Automated tracking of due dates and fines
· Better inventory management

· Improved member satisfaction

· Reduced operational costs

· Access to valuable usage statistics

2. Entity Relationship Diagram:

ERD (Entity Relationship Diagram) for a bank management system is a graphical representation of the entities, relationship s and attributes involved in the system. It is a useful tool for designing, implementing, and maintaining a database that supports the library operations.

2.1 Entities:

- Members
- Authors
- Bublishers
- Books
- Book_Authors
- Loans
- Fines

2.2 Attributes:

2.2.1Member: Member_id (PK), First_name,last_name,Email,Phone,Address, membership_date,Is_active. ·
Member_ID (Primary Key) - Unique member identifier
First_name, Last_name - Member's full name
Email, Phone, Address - Contact information

Membership_date - When member joined
Is_active - Current membership status

2.2.2 Author: Author_id (PK),First_name,Last_name,BiO

· Author_ID (Primary Key) - Unique author identifier

· First_name, Last_name - Author's name
· Bio - Author biography and details

2.2.3 Publisher: publisher_id (PK),publisher_name,address,phone.

· Publisher_ID (Primary Key) - Unique publisher identifier
· Publisher_name - Company name
· Address, Phone - Contact information

2.2.4 Book: Book_id (PK),Title,Isbn,Publication_year,Publisher_id (FK),Shelf_location,Total_copies,Available_copies

· Book_ID (Primary Key) - Unique book identifier
· Title - Book title
· ISBN - International Standard Book Number
· Publication_year - Year published
· Publisher_ID (Foreign Key) - Links to Publishers table
· Shelf_location - Physical location in library
· Total_copies - Number of copies owned
· Available_copies - Copies currently available

2.2.5 Book_Author: (Junction Table for Many-to-Many Relationship) book_id (PK, FK),Author_id (PK, FK)

2.2.6 Loan: Loan_id (PK),Member_id (FK),Book_id

(FK),Loan_date,Due_date,Return_date,Status (e.g., 'On Loan', 'Returned', 'Overdue')

· Loan_ID (Primary Key) - Unique loan identifier
· Member_ID (Foreign Key) - Links to Members table
· Book_ID (Foreign Key) - Links to Books table
· Loan_date - When book was borrowed
· Due_date - When book should be returned
· Return_date - Actual return date
· Status - Current loan status (On Loan, Returned, Overdue)

2.2.7 Fines: Fine_id (PK),Loan_id (FK),Fine_amount,Issue_date,Paid_date,Status (e.g., 'Issued', 'Paid')

· Fine_ID (Primary Key) - Unique fine identifier
· Loan_ID (Foreign Key) - Links to Loans table
· Fine_amount - Money owed
· Issue_date - When fine was created
· Paid_date - When fine was paid
· Status - Fine status (Issued, PaiD)

3. Relationship:

3.1 One-to-Many Relationships:

1.One publisher can publish many books((1:N) one-to-many relationship). Implemented via publisher_id foreign key in Books table.

2. One member can have multiple loans ((1:N) one-to-many relationship). Implemented via member_id foreign key in Loans table.

3.One book can be loaned multiple times over its lifetime(1:N)
Implemented via book_id foreign key in Loans table

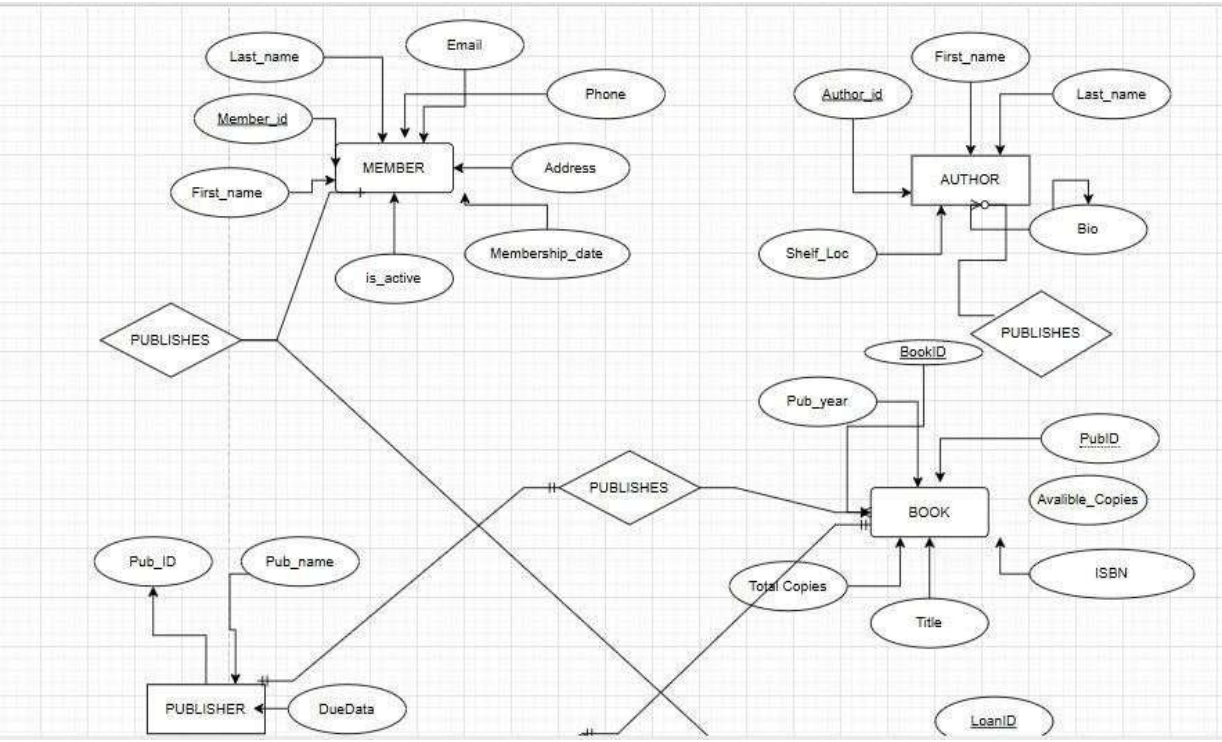
4.One loan can generate at most one fine
Implemented via loan_id unique foreign key in Fines table

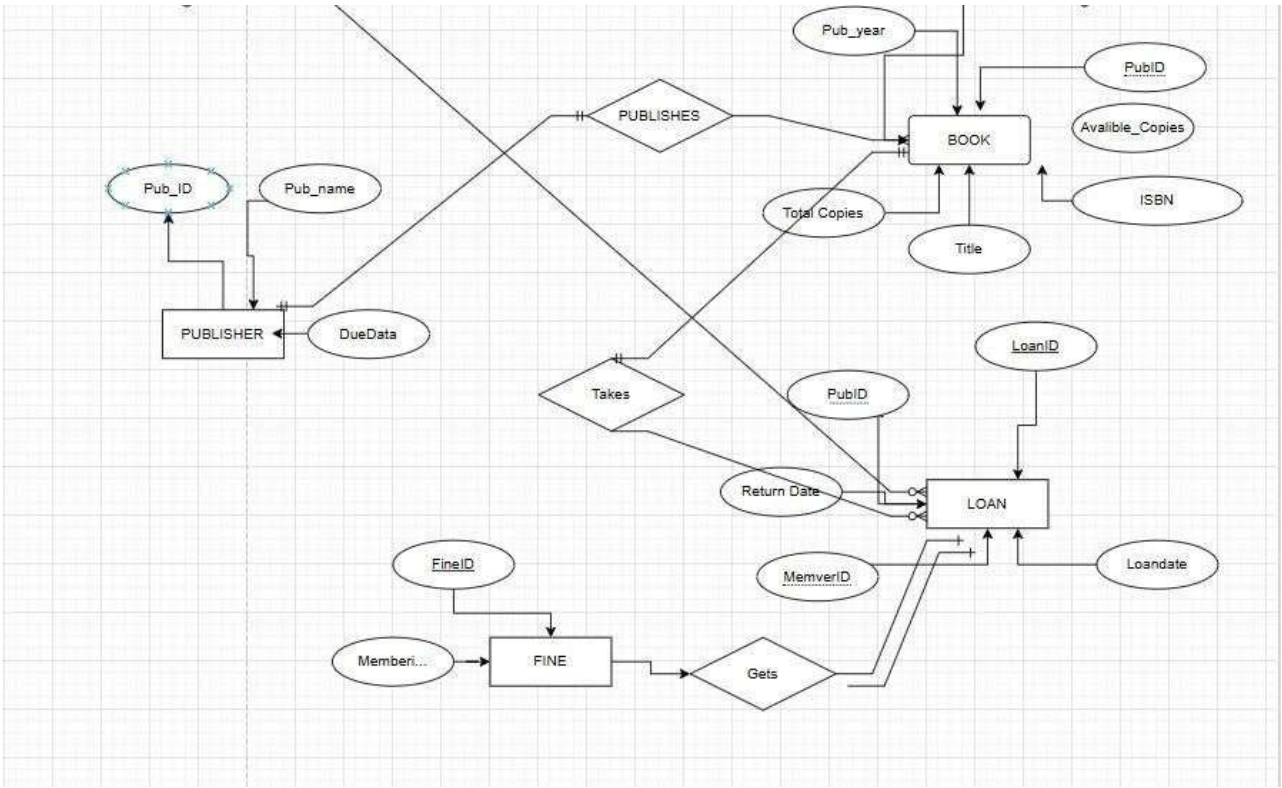
3.2 Many-to-Many Relationships:

1.One book can have multiple authors (M:N)
2.One author can write multiple books(M;N)

Resolved through *Book_Authors junction table
Junction table contains composite primary key (book_id, author_id)

4.Entity Relationship Diagram ERD (DRAW.IO)

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BOOK
Avalible_Copies
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Total Copies
ISBN
Title
PUBLISHER
DueData
LoanID
Takes
PubID
Return Date
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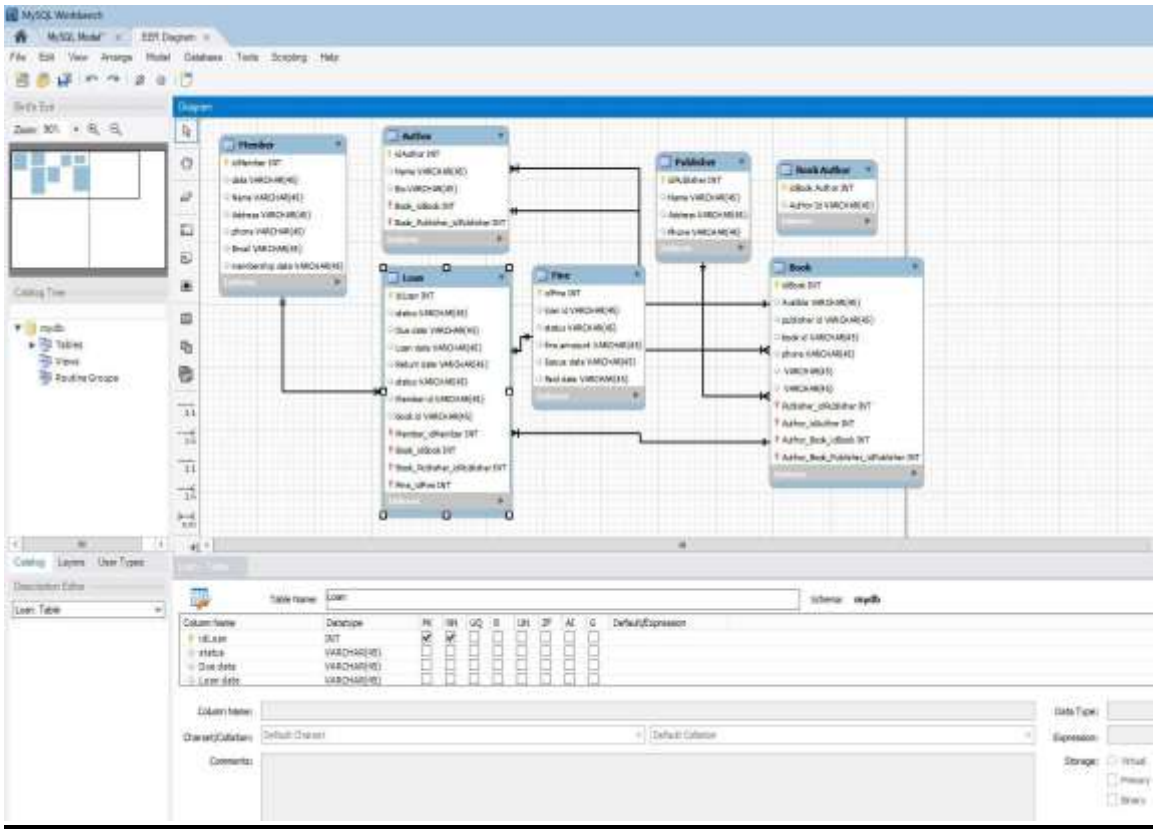
5.MYSQL Tables:



6.1)Member:

Table Name: Member

Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G
Name	VARCHAR(45)	☐	☐	☐	☐	☐	☐	☐	☐
Address	VARCHAR(45)	☐	☐	☐	☐	☐	☐	☐	☐
phone	VARCHAR(45)	☐	☐	☐	☐	☐	☐	☐	☐
Email	VARCHAR(45)	☐	☐	☐	☐	☐	☐	☐	☐

Table Name: Member

Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G	Defaul
phone	VARCHAR(45)	☐	☐	☐	☐	☐	☐	☐	☐	
Email	VARCHAR(45)	☐	☐	☐	☐	☐	☐	☐	☐	
membership date	VARCHAR(45)	☐	☐	☐	☐	☐	☐	☐	☐	
		☐	☐	☐	☐	☐	☐	☐	☐	

6.2 Author:

Table Name: Author

Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G	Default/
idAuthor	INT	☑	☑	☐	☐	☐	☐	☐	☐	
Name	VARCHAR(45)	☐	☐	☐	☐	☐	☐	☐	☐	
Bio	VARCHAR(45)	☐	☐	☐	☐	☐	☐	☐	☐	
Book_idBook	INT	☑	☑	☐	☐	☐	☐	☐	☐	

Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G	Default/Expressi
Book_idBook	INT	☑	☑	☐	☐	☐	☐	☐	☐	
Book_Publisher_idPublisher	INT	☑	☑	☐	☐	☐	☐	☐	☐	
		☐	☐	☐	☐	☐	☐	☐	☐	

6.3) Publisher:

Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G	Default/Expre
idPublisher	INT	☑	☑	☐	☐	☐	☐	☐	☐	
Name	VARCHAR(45)	☐	☐	☐	☐	☐	☐	☐	☐	
Address	VARCHAR(45)	☐	☐	☐	☐	☐	☐	☐	☐	
Phone	VARCHAR(45)	☐	☐	☐	☐	☐	☐	☐	☐	

Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G	Defau
Address	VARCHAR(45)	☐	☐	☐	☐	☐	☐	☐	☐	
Phone	VARCHAR(45)	☐	☐	☐	☐	☐	☐	☐	☐	
		☐	☐	☐	☐	☐	☐	☐	☐	

6.4)Book:

Table Name: book

olumn Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G	Defa
publisher id	VARCHAR(45)	☐	☐	☐	☐	☐	☐	☐	☐	
book id	VARCHAR(45)	☐	☐	☐	☐	☐	☐	☐	☐	
phone	VARCHAR(45)	☐	☐	☐	☐	☐	☐	☐	☐	
	VARCHAR(45)	☐	☐	☐	☐	☐	☐	☐	☐	

Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G	Default/Expression
Publisher_idPublisher	INT	☑	☑	☐	☐	☐	☐	☐	☐	
Author_idAuthor	INT	☑	☑	☐	☐	☐	☐	☐	☐	
Author_Book_idBook	INT	☑	☑	☐	☐	☐	☐	☐	☐	
Author_Book_Publisher_idPu...	INT	☑	☑	☐	☐	☐	☐	☐	☐	

Column Name: Email

6.5)Book_Author:

Table Name: Book Author

Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G	Default/Expression
idBookAuthor	INT	☑	☑	☐	☐	☐	☐	☐	☐	
Author Id	VARCHAR(45)	☐	☐	☐	☐	☐	☐	☐	☐	
		☐	☐	☐	☐	☐	☐	☐	☐	

6.6)Loan:

Table Name: Loan

Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G	Default/Ex
idLoan	INT	☑	☑	☐	☐	☐	☐	☐	☐	
status	VARCHAR(45)	☐	☐	☐	☐	☐	☐	☐	☐	
Due date	VARCHAR(45)	☐	☐	☐	☐	☐	☐	☐	☐	
Loan date	VARCHAR(45)	☐	☐	☐	☐	☐	☐	☐	☐	

Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G	Default/Expression
Member_idMember	INT	☑	☑	☐	☐	☐	☐	☐	☐	
Book_idBook	INT	☑	☑	☐	☐	☐	☐	☐	☐	
Book_Publisher_idPublisher	INT	☑	☑	☐	☐	☐	☐	☐	☐	
Fine_idFine	INT	☑	☑	☐	☐	☐	☐	☐	☐	

Table Name: Loan

Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G	Default/E
Return date	VARCHAR(45)	☐	☐	☐	☐	☐	☐	☐	☐	
status	VARCHAR(45)	☐	☐	☐	☐	☐	☐	☐	☐	
Member id	VARCHAR(45)	☐	☐	☐	☐	☐	☐	☐	☐	
book id	VARCHAR(45)	☐	☐	☐	☐	☐	☐	☐	☐	

6.7) Fines:

Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G	De
idFine	INT	☑	☑	☐	☐	☐	☐	☐	☐	
loan id	VARCHAR(45)	☐	☐	☐	☐	☐	☐	☐	☐	
status	VARCHAR(45)	☐	☐	☐	☐	☐	☐	☐	☐	
fine amoount	VARCHAR(45)	☐	☐	☐	☐	☐	☐	☐	☐	

Table Name:

Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G	Default/Expression
fine amoount	VARCHAR(45)	☐	☐	☐	☐	☐	☐	☐	☐	
Isssue date	VARCHAR(45)	☐	☐	☐	☐	☐	☐	☐	☐	
Paid date	VARCHAR(45)	☐	☐	☐	☐	☐	☐	☐	☐	
		☐	☐	☐	☐	☐	☐	☐	☐	

Column Name: Email